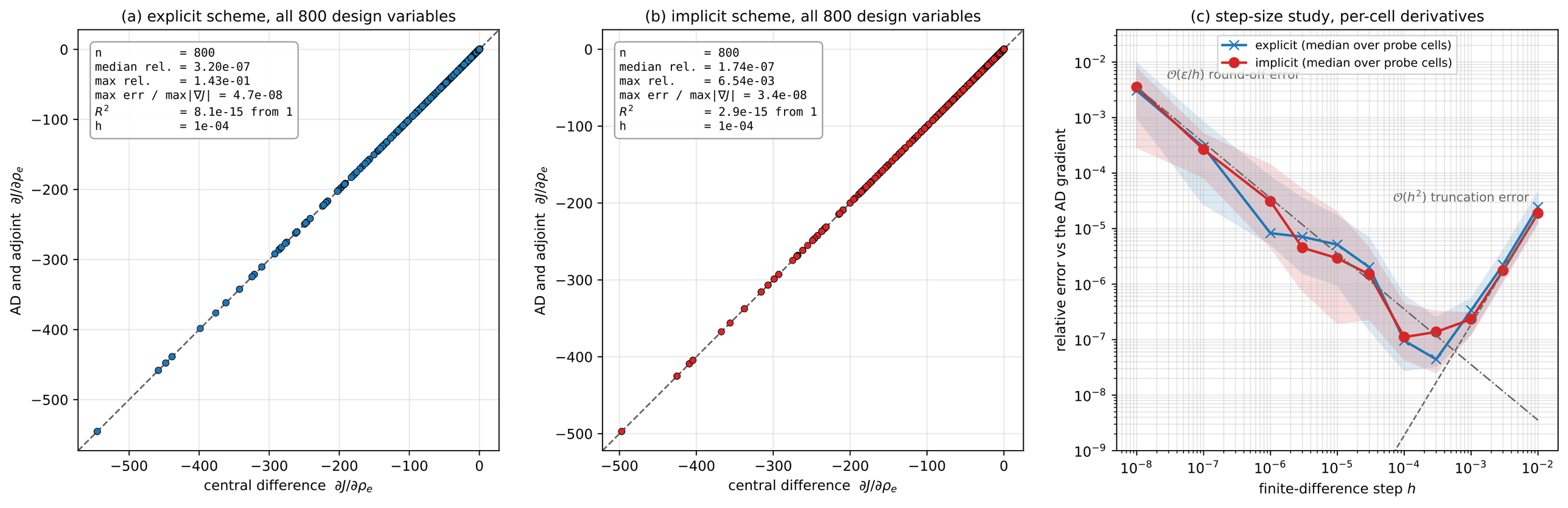

(a) explicit scheme, all 800 design variables
n = 800
median rel. = 3.20e-07
max rel. = 1.43e-01
max err / max|∇J| = 4.7e-08
R² = 8.1e-15 from 1
h = 1e-04
AD and adjoint ∂J/∂ρe
central difference ∂J/∂ρe
(b) implicit scheme, all 800 design variables
n = 800
median rel. = 1.74e-07
max rel. = 6.54e-03
max err / max|∇J| = 3.4e-08
R² = 2.9e-15 from 1
h = 1e-04
AD and adjoint ∂J/∂ρe
central difference ∂J/∂ρe
(c) step-size study, per-cell derivatives
explicit (median over probe cells)
implicit (median over probe cells)
O(ε/h) round-off error
O(h²) truncation error
relative error vs the AD gradient
finite-difference step h